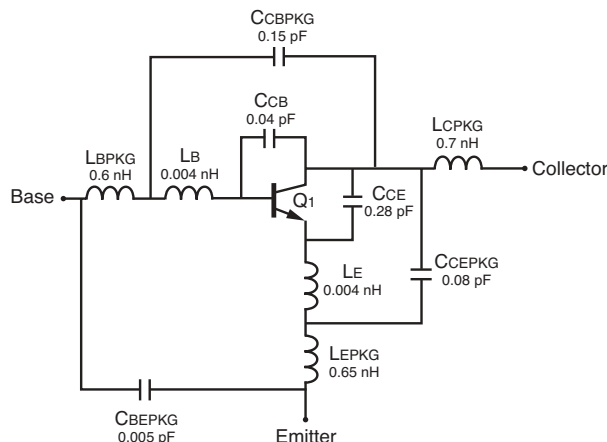


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	734.5e-18	MJC	0.122
BF	166.6	XCJC	0.1
NF	1.00	CJS	0
VAF	41	VJS	0.75
IKF	0.597	MJS	0
ISE	39.37e-15	FC	0.5
NE	2.258	TF	13e-12
BR	28.67	XTF	0.39
NR	1.000	VTF	0.668
VAR	2.541	ITF	0.06
IKR	23.22e-3	PTF	20
ISC	27.52e-18	TR	0
NC	2.0	EG	1.11
RE	1.7	XTB	0
RB	3.0	XTI	3
RBM	1.0	KF*	0
IRB	759e-6	AF*	1
RC	4.0		
CJE	2.51e-12		
VJE	0.887		
MJE	0.332		
CJC	498.2e-15		
VJC	0.367		

(1) Gummel-Poon Model

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.

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DATA SUBJECT TO CHANGE WITHOUT NOTICE

ADDITIONAL PARAMETERS

Parameters	NE851M03
CCB	0.04 pF
CCE	0.28 pF
LB	0.004 nH
LE	0.004 nH
CCBPKG	0.15 pF
CCEPKG	0.08 pF
CBEPKG	0.005 pF
LBPKG	0.6 nH
LCPKG	0.7 nH
LEPKG	0.65 nH

AF and KF are 1/f noise parameters and are bias dependent. The appropriate values for the 1/f noise parameters (AF and KF) shall be chosen from the table below, according to the desired current range.

	Ic = 5 mA	Ic = 10 mA	Ic = 15 mA
AF	1.40	2.551	2.626
KF	4.547e-15	855.6e-12	1.735e-9

For a better understanding on AF and KF parameters, please refer to AN1026.

MODEL TEST CONDITIONS

Frequency: 0.1 to 6.0 GHz

Bias: VCE = 1.5 V, Ic = 1 mA to 9 mA

Date: 09/2003